ReviewIQ

Aspect-Based Customer Review Analytics with Structured Outputs

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Problem

- Thousands of free-text reviews; star ratings say **how much**, not **why**.
- Goal: validated structured records — overall sentiment, aspects with evidence, complaint categories — and per-product summaries.
- Compare prompting techniques for sentiment and multi-label classification on public e-commerce data.
- Constraint: run on a student laptop, free, with no review data leaving the machine.

Solution

- Local LLMs through **Ollama** (qwen2.5:3b extraction, llama3.2 judge) — no API key, no cloud.
- **Pydantic** schema + evidence grounding; **automatic repair** of malformed outputs; controlled failures.
- Three prompting strategies behind one interface; resumable, cached experiments with honest status labels.
- Extract → Aggregate → Summarise chain with cited summaries and an LLM-as-a-Judge; six-page Streamlit dashboard.

Architecture

reviews → **EXTRACT** (prompt → local LLM → parse → validate + grounding → repair)

→ **AGGREGATE** (pandas statistics, evidence quotes)

→ **SUMMARISE** (cited summary, ids validated)

→ **JUDGE** (separate model, rubric 1–5 + deterministic checks)

LangChain ChatOllama · 127.0.0.1:11434 only · live and mock outputs physically separated

Three deliverables

D1 Structured extraction
Pydantic schema, LangChain parser instructions, JSON/CSV export, repair loop, resumable cache, outage-safe batches.

D2 Prompt comparison
Zero-shot · Few-shot (synthetic examples) · Chain-of-Thought (silent procedure), same model/data/settings; macro-F1 primary.

D3 Prompt chain
Extract → aggregate → summarise per product, every point cites review ids; LLM-as-a-Judge (faithfulness, coverage, consistency, hallucination) + deterministic checks.

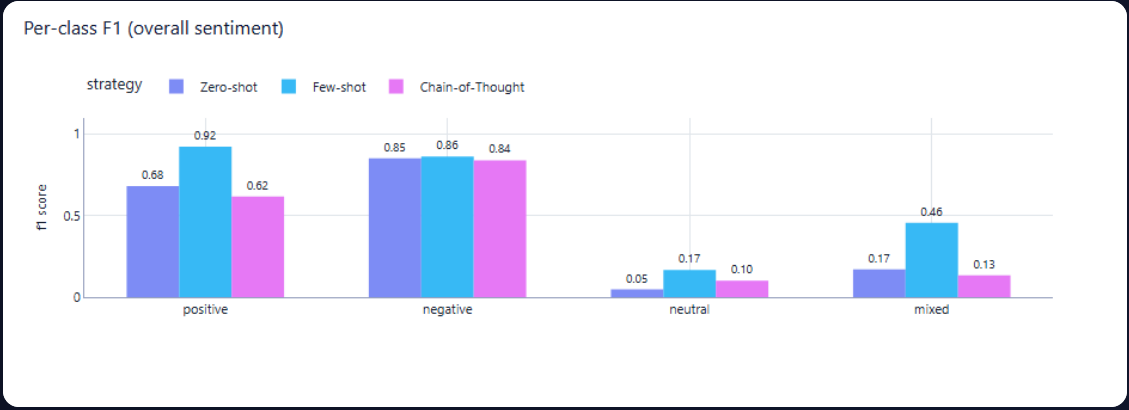
D2 results — qwen2.5:3b, 299 reviews

strategy	macro F1	accuracy	coverage	validity	repairs	s / review
Zero-shot	0.438	0.612	1.000	0.943	26	41.2
Few-shot	0.603	0.813	1.000	0.953	17	53.8
Chain-of-Thought	0.423	0.575	1.000	0.933	25	47.5

Best macro-F1: **Few-shot** (0.603). Status: **complete**.

Reference labels: 1 human + 298 AI-assisted — scores measure agreement with these labels.

Per-class F1 and confusion





D3 results — product summaries and judge

faithfulness	coverage	consistency	hallucination
3.80	**3.80**	**3.00**	**1.40**
mean of 5 (1–5)	mean of 5 (1–5)	mean of 5 (1–5)	mean of 5 (1–5)

15 products summarised · **5 judged, 10 judge call(s) failed** (invalid judge output; no score) · citation-polarity rate 0.96 · unsupported claims flagged: 1

Scope: the 299-review evaluation sample (285 extracted reviews), not the full 1,500. Judge = llama3.2 (3B); scores are model opinions, audited with deterministic checks — not objective truth.

Does the judge catch a bad summary?

Negative control: each summary with a valid original judgement (5 of 15) copied and corrupted (inverted opinion + fabricated %, invented feature, complaints removed), judged by the same model; 3 pair(s) scored, 2 corrupted judgement(s) failed.

mean score — original	mean score — corrupted	corrupted scored lower (mean)	corrupted faithfulness lower
3.17	2.92	3/3	0/3

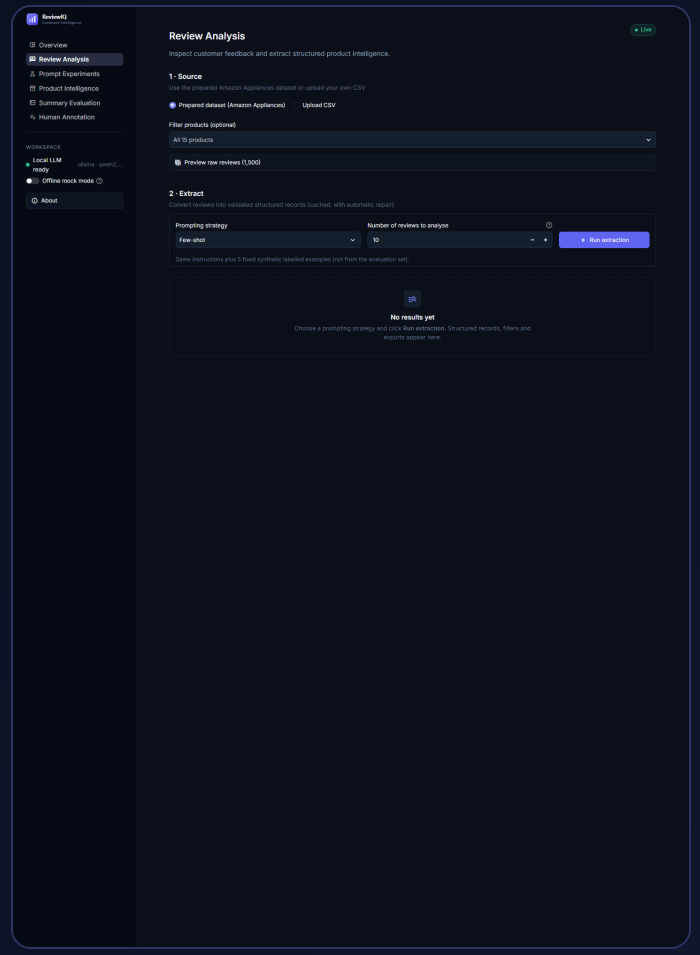
Invented feature flagged in 0/3 · fabricated percentage flagged in 3/3 · equal scores in 0/3. Too few pairs for a conclusion.

Demo — Overview



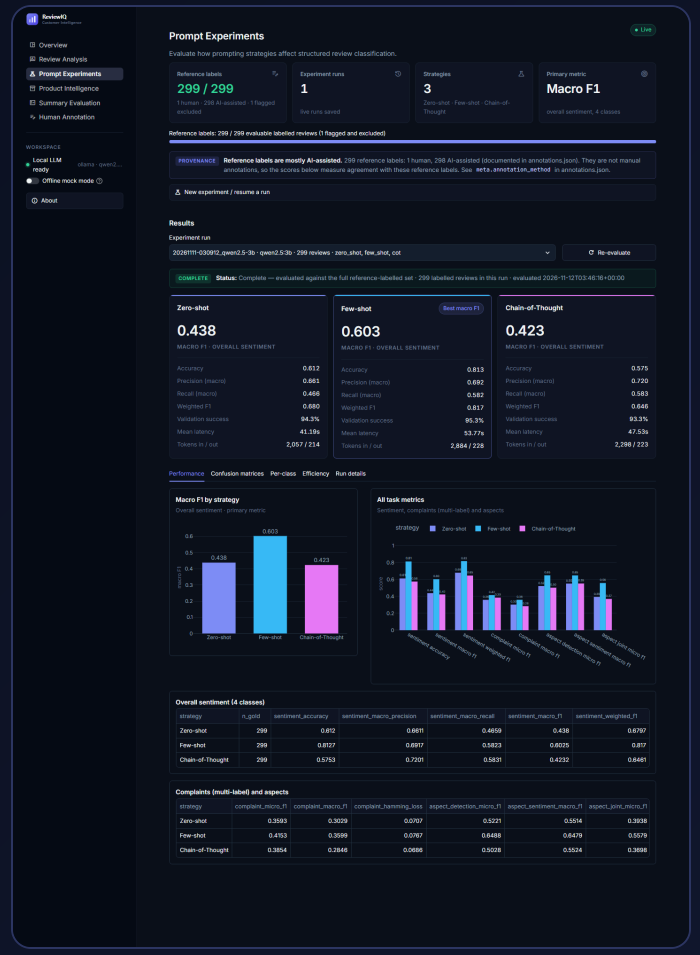
Overview — dataset size, deliverable status and processed reviews

Demo — Review Analysis



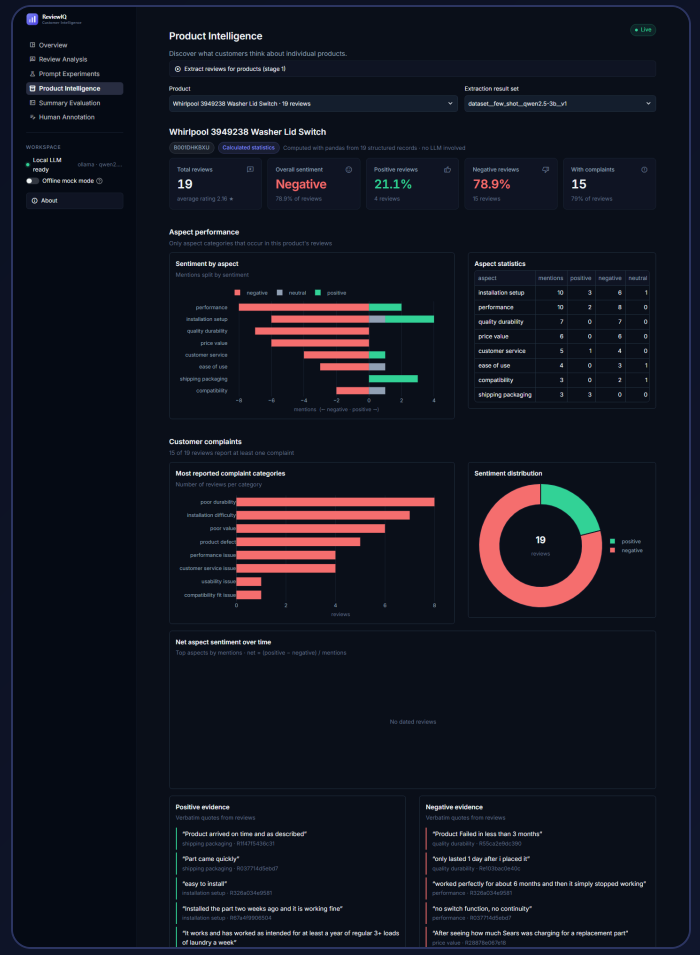
Review Analysis — the extraction page in its initial state (data source, prompting strategy and run controls); no extraction was run in this browser session, so no records are shown

Demo — Prompt Experiments



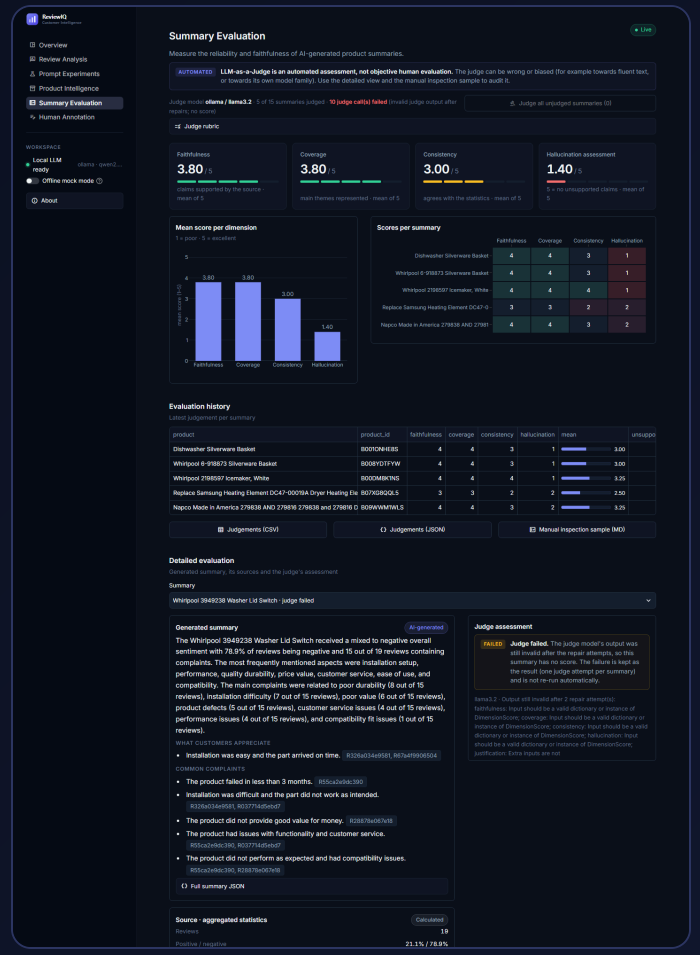
Prompt Experiments — measured comparison of the three strategies

Demo — Product Intelligence



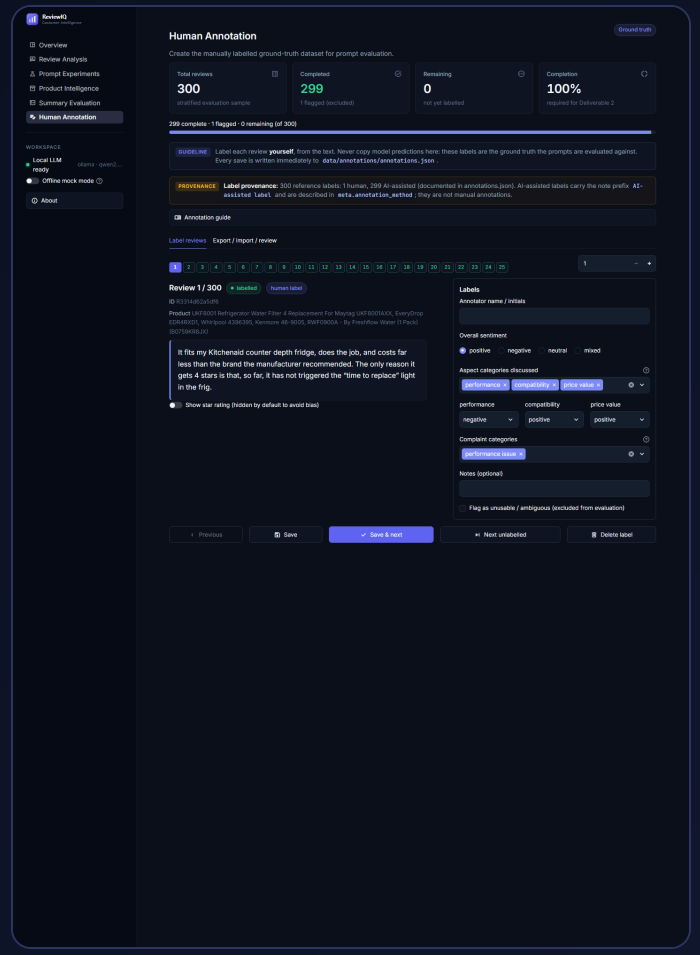
Product Intelligence — calculated statistics, aspect performance, complaint categories and evidence quotes for one product; the generated summary lower on the page is outside the captured area

Demo — Summary Evaluation



Summary Evaluation — LLM-as-a-Judge scores and deterministic checks

Demo — Human Annotation



Human Annotation — reference labels with provenance

Demo flow

1. Overview: dataset (1,500 reviews, 15 products) and deliverable status.
2. Review Analysis: run one review live, show JSON, evidence and repair log.
3. Prompt Experiments: comparison table, macro-F1, confusion matrices, provenance note.
4. Product Intelligence: aggregated aspects/complaints and the cited summary.
5. Summary Evaluation: judge scores, unsupported claims, deterministic checks.
6. Human Annotation: reference labels with human / AI-assisted badges.

Limitations

- Reference labels mostly AI-assisted (1 human) — agreement, not independent ground truth.
- Small local models: 3B misses aspects/complaints; 7B too slow on a 4 GB GPU for the 299-review evaluation (smoke test only).
- Occasional runaway generation, handled by an output cap + repair.
- Category-level aspect matching; rating-stratified sample; 3B judge not human-calibrated.

Conclusion

- A complete, local, reproducible pipeline: validated extraction, fair prompt comparison, cited summaries, auditable judge.
- Judge reliability: 5/15 valid judge outputs (10 failed); faithfulness lower for corrupted summaries in 0/3 scored pairs — not a reliable quality measure.
- Validation + repair make a small local model usable for structured output; the metrics show where 3B models fall short.
- Next: human re-annotation, constrained JSON decoding, larger local model, human-calibrated judging.

Thank you